

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0567

Somerville 138 kV
February 28, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0567. The Interconnection Customer proposes a 156.0 MW wind facility interconnecting at the Somerville 138 kV substation in the Pennsylvania area. The request was received on January 05, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 291 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,783,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RECLAIM-CHASE HILL 138 kV	BURWELL-P.T.S 345 kV line	174.4	210	22.9
RECLAIM-CENTERVILLE RD 138 kV	BURWELL-P.T.S 345 kV	166.6	233	22.9
CENTERVILLE RD-SAN CARLOS 138 kV	P.T.S-BURWELL 345 kV Line	164.1	244	22.9
SAN CARLOS-SOMERVILLE 138 kV	BURWELL-P.T.S 345 kV Line	162.6	252	22.9
KILARC SUBSTATION-THERMAS 138 kV	CONEMAUGH HYDRO PLANT-THERMAS 138 kV L	121.4	177	5.9
WAHIAWA-N ALLEGHENY WINDPOWER PROJECT 345 kV	SUNRISE CTR-MERRET 500 kV Line	120.4	1,447	5.8
BLACKROCK-EKLUTNA SUBSTATION 345 kV	CALDWELL-PINEY CRK PROJECT 345 kV	119.8	1,136	5.2
CHASE HILL-EMILY 3 138 kV	P.T.S-BURWELL 345 kV	113.1	211	10.3
ST CROIX FALLS-EMILY 3 138 kV Line	BURWELL-P.T.S 345 kV	113.0	199	10.3
PLACER-STINSON 138 kV	SOUTHEAST VICKSBURG-ALLEGHENY ENERGY UNITS 3 4 & 5 500 kV	112.8	331	6.1
EKLUTNA-ALLEGHENY HYDRO NO 8 345 kV	CALDWELL SUBSTATION-PINEY CREEK PROJECT 345 kV	112.6	1,188	5.3
SUNRISE CENTER-LUTESVILLE 345 kV Line	CAYUCOS-SUNRISE CENTER 500 kV	112.3	1,629	5.1
WILSON 1&2 345/138kV Autotransformer	SAN CALOS-SOMERVILLE 138 kV Line	112.2	374	6.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PACOLET DIESEL GENERATION FACILITY-STINSON 138 kV L	ALEGHENY ENERGY UNITS 3 4 & 5-SOUTHEAST VICKSBURG 500 kV	112.0	226	6.3
THERMAS-WILSON 1&2 138 kV	HILLTOP-PIEY CREEK PROJECT 500 kV line	111.1	323	7.3
W CAMPUS-EDES TAP 2 345 kV L	TALENENERGY MARTINS CRK LLC WILLIAMSPORT-CAYUCOS 500 kV	109.1	1,639	5.5
EDES TAP 2-LUTESVILLE 345 kV	TALENENERGY MARTINS CREEK LLC WILLIAMSPORT-CAYUCOS SUBSTATION 500 kV	106.7	1,642	5.3
GROVE-PACOLET DIESEL GNERATION FACILITY 138 kV	ALLEGHENY ENERGY UNITS 3 4 & 5-SOUTHEAST VICKSBURG SUBSTATION 500 kV line	105.3	391	8.6
N WAVERLY-GROVE 138 kV line	ALLEGHENY ENERGY UNITS 3 4 & 5-SOUTHEAST VICKSBURG 500 kV Line	104.7	405	8.6
FIRSTENERGY SPRINGDALE POWER STATION-N WAVERLY 138 kV line	SOUTHEAST VICKSBURG-ALLEGHENY ENERGY UNITS 3 4 & 5 500 kV	104.5	408	8.6
NORTH GATE RIVERSIDE ALTE-P.T.S 138 kV	P.T.S-BURWELL 345 kV	102.3	245	10.9
N GATE RIVERSID ALTE SUBSTATION-DELCO 138 kV	P.T.S-BURWELL 345 kV	101.8	233	10.9
WILMERDING-DELCO 138 kV	BURWELL-P.T.S 345 kV line	101.5	221	10.9
ROBISON PARK-STATE ST 230 kV	DEKALB SUBSTATION-RICHLAND 500 kV	101.0	1,124	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHASE HILL-RECLAIM 138 kV	Rebuild line with higher-capacity conductor	\$2,360,000	43% (\$1,007,000)
RECLAIM-CENTERVILLE RD 138 kV	Rebuild line with higher-capacity conductor	\$2,208,000	43% (\$942,000)
CENTERVILLE RD-SAN CARLOS 138 kV	Rebuild line with higher-capacity conductor	\$599,000	43% (\$256,000)
SAN CARLOS-SOMERVILLE 138 kV	Rebuild line with higher-capacity conductor	\$1,519,000	43% (\$648,000)
KILARC-THERMAS 138 kV	Rebuild line with higher-capacity conductor	\$4,421,000	53% (\$2,361,000)
WAHIAWA-N ALLEGHENY WINDPOWER PROJECT 345 kV	Rebuild line with higher-capacity conductor	\$45,982,000	3% (\$1,468,000)
EKLUTNA-BLACKROCK 345 kV	Rebuild line with higher-capacity conductor	\$55,077,000	16% (\$8,687,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHASE HILL-EMILY 3 138 kV L	Rebuild line with higher-capacity conductor	\$3,185,000	49% (\$1,573,000)
ST CROIX FALLS-EMILY 3 138 kV Line	Rebuild line with higher-capacity conductor	\$2,061,000	49% (\$1,018,000)
PLACER-STINSON 138 kV	Rebuild line with higher-capacity conductor	\$3,269,000	30% (\$984,000)
EKLUTNA-ALLEGHENY HYDRO NO 8 345 kV	Rebuild line with higher-capacity conductor	\$72,578,000	19% (\$13,821,000)
SUNRISE CENTER-LUTESVILLE 345 kV Line	Rebuild line with higher-capacity conductor	\$44,364,000	3% (\$1,248,000)
WILSON 1&2 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$18,242,000	23% (\$4,270,000)
PACOLET DIESEL GENERATION FACILITY-STINSON 138 kV L	Rebuild line with higher-capacity conductor	\$2,626,000	26% (\$684,000)
THERMAS-WILON 1&2 138 kV	Rebuild line with higher-capacity conductor	\$3,432,000	54% (\$1,846,000)
W CAMPUS-EDES TAP 2 345 kV L	Reconductor line	\$14,831,000	3% (\$435,000)
EDES TAP 2-LUTESVILLE 345 kV	Reconductor line	\$15,720,000	3% (\$478,000)
GROVE-PACOLET DIESEL GENERATION FACILITY 138 kV	Reconductor line	\$2,408,000	25% (\$597,000)
NORTH WAVERLY-GROVE 138 kV line	Reconductor line	\$2,771,000	25% (\$687,000)
FIRSTENERGY SPRINGDALE POWER STATION-N WAVERLY 138 kV line	Reconductor line	\$1,279,000	25% (\$317,000)
N GATE RIVERSIDE ALTE-P.T.S 138 kV	Reconductor line	\$417,000	100% (\$417,000)
N GATE RIVERSID ALTE SUBSTATION-DELCO 138 kV	Reconductor line	\$475,000	100% (\$475,000)
DELCO-WILMERDING 138 kV Line	Reconductor line	\$436,000	100% (\$436,000)
ROBISON PARK-STATE ST 230 kV	Reconductor line	\$3,827,000	3% (\$128,000)
	Total allocated to project		\$44,783,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.